LESSON 01 / 18

Read a differential equation before solving it

A differential equation relates an unknown function to its derivatives. Its **order** is the highest derivative present. Its **degree** is the power of that highest derivative after the equation has been written as a polynomial in the derivatives, with no radical or transcendental operation applied to a derivative.

A solution is a function satisfying the equation on a stated **interval**. An initial-value problem, or IVP, also prescribes values such as $y(x_0) = y_0$. A general solution contains arbitrary constants; a particular solution results from determining them. Constants do not remove the need to state the domain.

The first four checks prevent most notation errors

Check	Ask	Trap
Order	Highest derivative?	Largest power instead
Degree	Power of that derivative?	Ignoring radicals
Solution	Identity after substitution?	One-point check only
Interval	Where is it defined?	Crossing a pole or domain edge

READ THE FIGURE

Verification is an identity on an interval, not a numerical coincidence at a single x-value.

WORKED EXAMPLE

S1 and Q2: verify rather than solve

For $y = e^{2x}$,

$$\frac{dy}{dx} = 2e^{2x} = 2y, \qquad \frac{dy}{dx} - 2y = 0.$$

The equality holds for every real x, so e^{2x} is a solution on $(-\infty, \infty)$. More generally, $y = Ce^{2x}$ solves the same equation; the initial condition $y(0) = 1$ selects $C = 1$.

WORKED EXAMPLE

PI and QI: order is not the largest visible power

In

$$\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^4 + y = 0,$$

the highest derivative is y''', so the order is 3. Its exponent is 1, so the degree is 1. The exponent 4 belongs to the lower derivative y' and does not determine either answer.

Two practical habits

- **Verify by differentiation:** compute the required derivatives, substitute, and simplify to zero or to the prescribed right-hand side.
- **Carry the interval:** formulas containing $\ln x$, $1/x$, or a denominator may represent different solution branches on different intervals.

COMMON TRAP

A formula can solve an equation without containing every solution

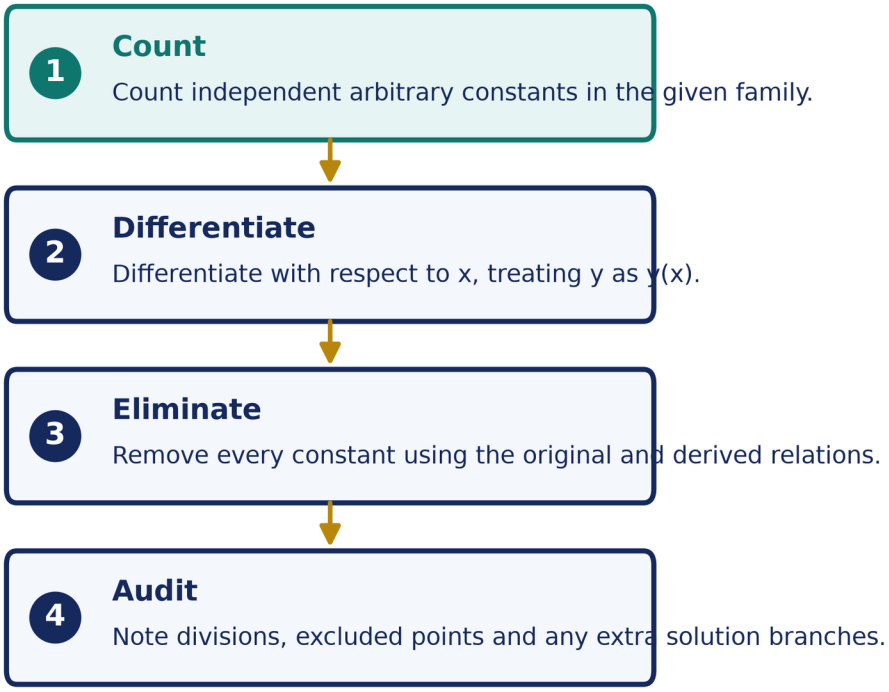
A finite choice of an arbitrary constant cannot always recover an equilibrium or singular solution. Keep track of every division by a quantity that might be zero; the Bernoulli, Lagrange and Riccati lessons apply this audit explicitly.

LESSON 02 / 18

Differentiate enough times, then eliminate

A family with n independent arbitrary constants usually requires n differentiations and leads to an equation of order n. The goal is not merely to differentiate: every arbitrary constant must be eliminated. The resulting differential equation can have additional solutions, so formation gives a necessary equation for the family, not always an identical solution set.

From a family of curves to a differential equation



READ THE FIGURE
One constant normally means first order; two independent constants normally mean second order.

WORKED EXAMPLE

S2 and Q3: the family y = cx + c squared

For $y = cx + c^2$, write $p = y'$. Differentiation gives $p = c$. Substitution in the original family yields

$$y = xy' + (y')^2.$$

This is a Clairaut equation. Its singular solution $y = -x^2/4$ is not one of the original straight lines; Extension II explains why the differential equation has that extra branch.

WORKED EXAMPLE

P2 and Q4: circles through the origin

The circles $(x-a)^2+y^2=a^2$ satisfy

$$x^2+y^2=2ax.$$

Implicit differentiation gives $2x+2yy'=2a$. Eliminating a on intervals where $x\neq 0$ gives

$$2x+2yy'=\frac{x^2+y^2}{x} \quad\Longrightarrow\quad 2xyy'=y^2-x^2.$$

The final polynomial form is the required formation result. At the origin it carries no slope information, so do not infer a unique tangent there.

WORKED EXAMPLE

Q41: two constants demand a second-order equation

For $y=c_1e^{x}+c_2e^{-x}$,

$$y'=c_1e^{x}-c_2e^{-x}, \qquad y''=c_1e^{x}+c_2e^{-x}=y.$$

Thus the required equation is $y''-y=0$. Conversely, its characteristic roots are 1 and -1, so the displayed family is its full real general solution.

COMMON TRAP

Do not differentiate without using the original relation

Differentiation alone leaves the constants in place. Always return to the family equation and eliminate them; otherwise the order may look right while the formation task remains unfinished.

LESSON 03 / 18

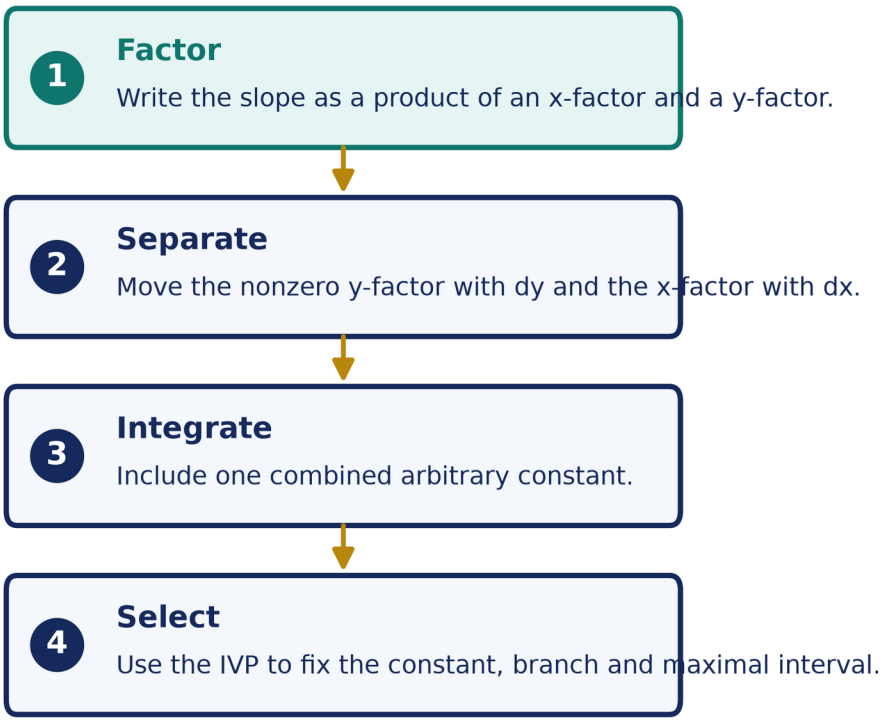
Put each variable on one side

For $y' = f(x)g(y)$, separation on a region where $g(y) \neq 0$ gives

$$\frac{dy}{g(y)} = f(x)\,dx.$$

Integrate both sides, apply the initial condition, and only then choose a branch. Constant solutions with $g(y) = 0$ must be checked separately before dividing.

A safe separation routine



READ THE FIGURE
If a division assumes y is nonzero, test the zero function or equilibrium separately.

WORKED EXAMPLE

S3 and Q5: choose the positive square-root branch

From $y' = x/y$,

$$y\,dy = x\,dx \quad \Longrightarrow \quad y^2 = x^2 + C.$$

Since $y(0) = 1$, $C = 1$. Continuity from the positive initial value selects

$$y = \sqrt{x^2 + 1}, \qquad y(1) = \sqrt{2}.$$

The negative square root belongs to a different initial condition, not to this IVP. This solution is defined on $(-\infty, \infty)$.

WORKED EXAMPLE

S4/Q6 and P3/Q7: exponential separation

For $y' = y$, $y(0) = 1$, separation for $y \neq 0$ gives $\ln|y| = x + C$, hence $y = e^x$ and $y(1) = e$. The zero function is another solution of the general equation but fails this IVP.

For $y' = 2xy$, $y(0) = 3$,

$$\frac{dy}{y} = 2x\,dx \quad \Longrightarrow \quad y = 3e^{x^2}, \qquad y(1) = 3e.$$

Both IVP solutions remain on the positive branch for all real x.

WORKED EXAMPLE

P4 and Q8: separate an exponential of a difference

Since $e^{x-y} = e^x e^{-y}$,

$$e^y\,dy = e^x\,dx \quad \Longrightarrow \quad e^y = e^x + C.$$

The condition $y(0) = 0$ gives $1 = 1 + C$, so $C = 0$ and $y = x$ on $(-\infty, \infty)$. Direct verification is immediate: $y' = 1 = e^{x-y}$.

COMMON TRAP

The constant belongs after integration, not after branch selection

Adding a new constant after taking a square root or exponential usually describes a different family. Carry one constant through the algebra and let the initial condition determine it.

LESSON 04 / 18

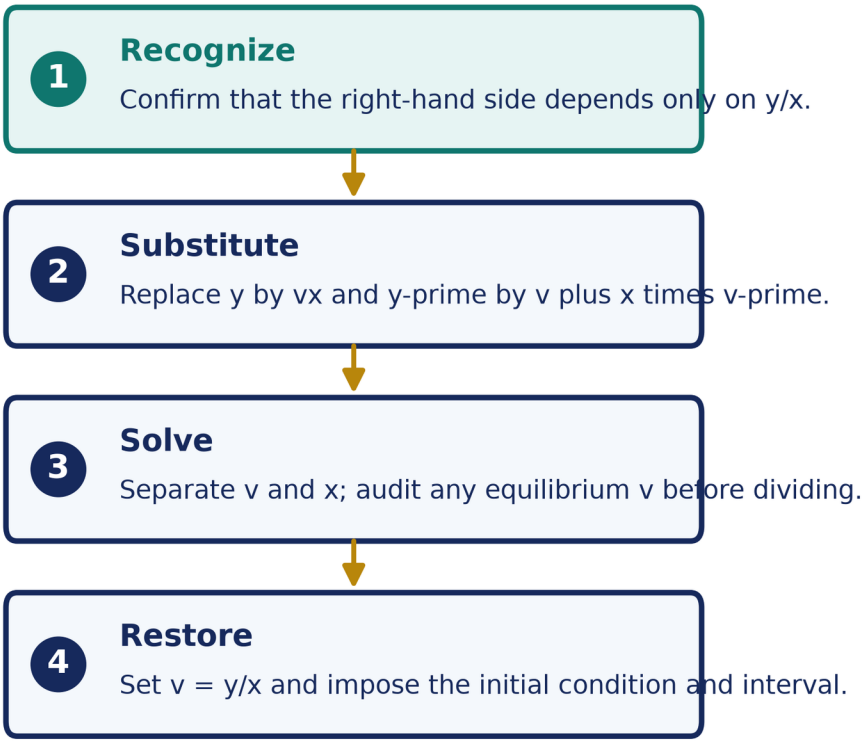
Use the ratio, not the word homogeneous alone

Here homogeneous means $y' = F(y/x)$ on an interval not containing $x = 0$. Put

$$y = vx, \qquad y' = v + xv'.$$

The v terms often cancel, leaving a separable equation in v and x. Return to $y = vx$ only after solving for v.

The substitution y = vx exposes the hidden separation



READ THE FIGURE

This meaning of homogeneous is different from a linear equation with zero right-hand side.

WORKED EXAMPLE

SS and Q9: immediate cancellation

For $y' = (x + y)/x = 1 + y/x$,

$$v + xv' = 1 + v \quad \Longrightarrow \quad xv' = 1.$$

Thus $v = \ln|x| + C$. With $y(1) = 0$, $v(1) = 0$ and $C = 0$; on $x > 0$,

$$y = x\ln x, \qquad y(e) = e.$$

The maximal IVP interval is $(0, \infty)$ because the original equation contains $1/x$.

WORKED EXAMPLE

PS and Q10: a finite blow-up boundary

For $y' = y/x + (y/x)^2$,

$$xv' = v^2 \quad \Longrightarrow \quad -\tfrac{1}{v} = \ln|x| + C$$

on a nonzero-v branch. The condition $y(1) = 1$ gives $v(1) = 1$ and $C = -1$. Therefore

$$y = \frac{x}{1 - \ln x}, \qquad y(2) = \frac{2}{1 - \ln 2}.$$

Starting at $x = 1$, the maximal interval is $(0, e)$: the denominator vanishes at $x = e$, while $x = 0$ is forbidden.

WORKED EXAMPLE

Q11: the radicand determines the lower endpoint

For $y' = (x^2 + y^2)/(xy) = x/y + y/x$,

$$v + xv' = v + \tfrac{1}{v} \quad \Longrightarrow \quad v\,dv = \tfrac{dx}{x}.$$

Hence $v^2 = 2\ln|x| + C$. Since $v(1) = 1$, $C = 1$, and the positive branch on $x > 0$ is

$$y = x\sqrt{2\ln x + 1}, \qquad y(e) = e\sqrt{3}.$$

The IVP interval is $(e^{-1/2}, \infty)$; at its lower endpoint both the radicand and y reach zero, so the original denominator xy fails.

COMMON TRAP

Never discard v = 0 without checking it

If separation divides by v, the constant ratio $v = 0$ may give $y = 0$ as an additional solution of the general equation. It does not affect the three nonzero IVPs above, but it matters when listing all branches.

LESSON 05 / 18

Multiply first, then recognize a product derivative

For

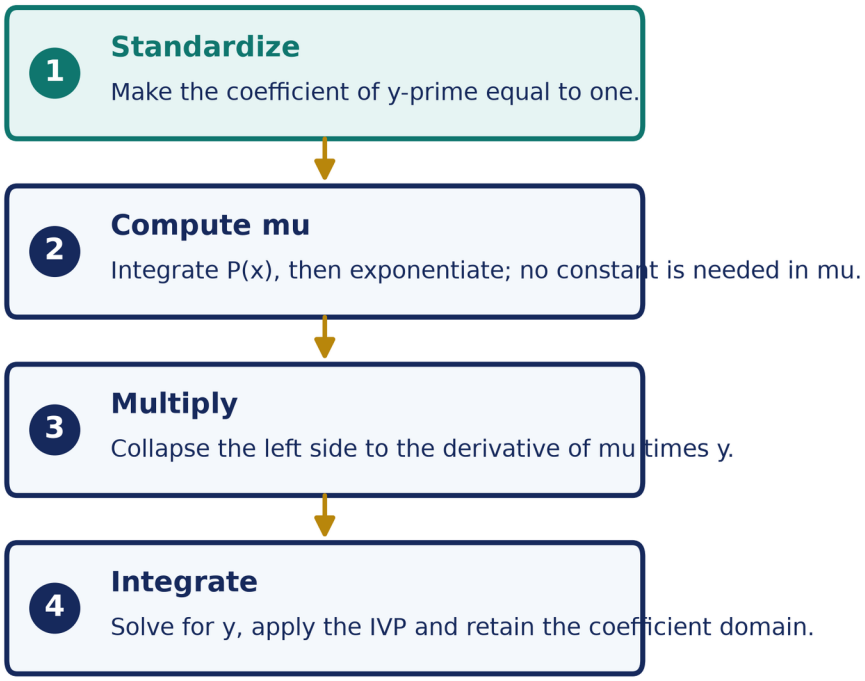
$$y' + P(x)y = Q(x),$$

choose an integrating factor $\mu = e^{\int P(x)dx}$. Then $\mu' = P\mu$, so

$$(\mu y)' = \mu Q, \qquad y = \tfrac{1}{\mu}\Big(\int \mu Q\, dx + C\Big).$$

Work on an interval where P and Q are continuous. The same μ works up to a nonzero constant multiple.

The integrating-factor method is product rule in reverse



READ THE FIGURE
Dividing by mu at the end is essential; stopping at mu y leaves the unknown still wrapped.

WORKED EXAMPLE

S6 and Q12: exponential forcing

For $y' + y = e^x$, $\mu = e^x$ and

$$(e^x y)' = e^{2x} \quad\Longrightarrow\quad y = \tfrac{1}{2}e^x + Ce^{-x}.$$

Since $y(0) = 0$, $C = -1/2$. Thus

$$y = \frac{e^x - e^{-x}}{2} = \sinh x, \qquad y(1) = \sinh 1.$$

WORKED EXAMPLE

S7/Q13 and Q16: polynomial and matching exponential forcing

For $y' + 2y = 4x$, $\mu = e^{2x}$. Integration, or the trial $ax + b$, gives

$$y = 2x - 1 + Ce^{-2x}.$$

The condition $y(0) = 1$ gives $C = 2$, so $y(1) = 1 + 2e^{-2}$.

For $y' + 2y = e^{-x}$, try Ae^{-x}: $(-A + 2A)e^{-x} = e^{-x}$, so $A = 1$. Hence $y = e^{-x} + Ce^{-2x}$. Since $y(0) = 1$, $C = 0$ and the IVP solution is simply $y = e^{-x}$ on $(-\infty, \infty)$.

WORKED EXAMPLE

P6 and Q14: keep x positive

For $y' + y/x = x$, the initial point $x = 1$ selects $x > 0$. Here $\mu = x$ and

$$(xy)' = x^2 \quad \Longrightarrow \quad xy = \tfrac{x^3}{3} + C.$$

Since $y(1) = 0$, $C = -1/3$. Therefore

$$y = \tfrac{x^2}{3} - \tfrac{1}{3x}, \qquad y(2) = \tfrac{7}{6},$$

with maximal IVP interval $(0, \infty)$.

COMMON TRAP

Standard form comes before the integrating factor

If the equation is $a(x)y' + b(x)y = r(x)$, divide by $a(x)$ first on an interval where it does not vanish. Using b/a incorrectly is the most common integrating-factor error.

LESSON 06 / 18

Bernoulli becomes linear after a power substitution

For $n \neq 0, 1$, the equation

$$y' + P(x)y = Q(x)y^n$$

has the substitution $u = y^{1-n}$. On a nonzero solution branch,

$$u' + (1-n)P(x)u = (1-n)Q(x).$$

Solve this linear equation and return to y. If $y = 0$ is defined in the original equation, check it separately before dividing by y^n.

Recognize the method from the right-hand side

Feature	Try	Audit
y^n forcing	u = y^(1-n)	Test y = 0 first
sin x forcing	A sin x + B cos x	Match both coefficients
exp(kx) forcing	IF or A exp(kx)	Check resonance
Pole in answer	Restrict x-interval	Never cross the pole

READ THE FIGURE
A substitution changes the algebra, not the obligation to verify the final function in the original equation.

WORKED EXAMPLE
S8 and Q15: Bernoulli with n = 2
For $y' + y = xy^2$, $y(0) = 1$, the zero equilibrium fails the IVP. Put $u = 1/y$. Dividing by y^2 gives

$$-u' + u = x \quad \Longrightarrow \quad u' - u = -x.$$

With integrating factor e^{-x},

$$(ue^{-x})' = -xe^{-x} \quad \Longrightarrow \quad u = x + 1 + Ce^{x}.$$

The initial condition gives $u(0) = 1$, hence $C = 0$. Therefore

$$y = \frac{1}{x+1}, \qquad y(1) = \frac{1}{2}.$$

Starting at $x = 0$, the maximal interval is $(-1, \infty)$, not all real numbers.

A sine forcing term needs both sine and cosine

For constant-coefficient linear equations, a trial $A\sin x + B\cos x$ accounts for the derivative interchanging sine and cosine.

WORKED EXAMPLE

Q28: solve y-prime + y = sin x

Substitute $y_p = A\sin x + B\cos x$ into $y' + y = \sin x$:

$$(A - B)\sin x + (A + B)\cos x = \sin x.$$

Thus $A - B = 1$ and $A + B = 0$, so $A = 1/2$, $B = -1/2$. The general solution is

$$y = \tfrac{1}{2}(\sin x - \cos x) + Ce^{-x}.$$

Since $y(0) = 0$, $C = 1/2$, and

$$y = \tfrac{1}{2}(\sin x - \cos x + e^{-x}), \qquad y\left(\tfrac{\pi}{2}\right) = \frac{1 + e^{-\pi/2}}{2}.$$

COMMON TRAP

A one-term sine trial is usually incomplete

The derivative of sine is cosine. Omitting the cosine trial forces an inconsistent coefficient comparison unless the equation has special structure.

LESSON 07 / 18

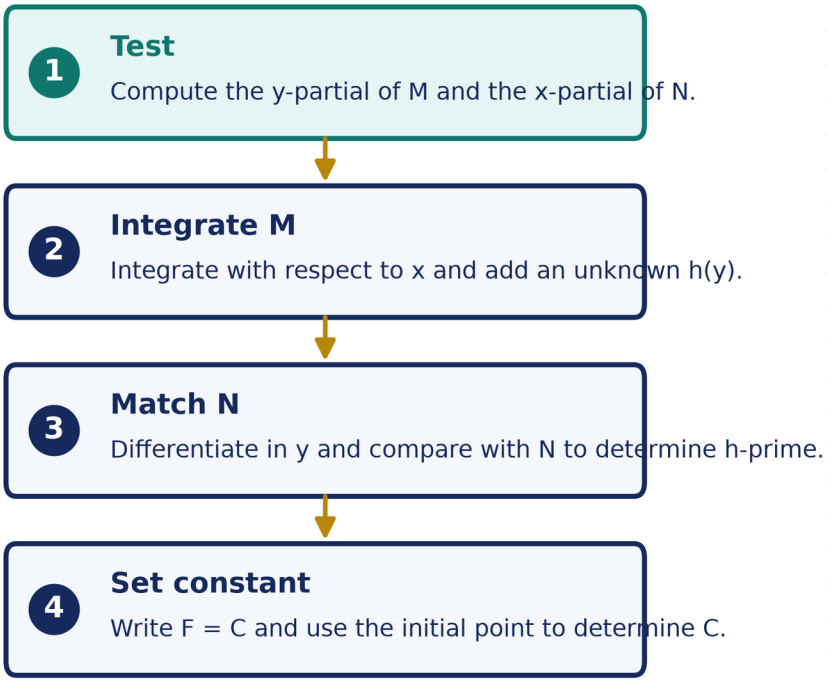
Find one function whose total differential vanishes

The differential form $M(x,y)dx + N(x,y)dy = 0$ is exact when there is a potential F with

$$F_x = M, \qquad F_y = N.$$

On a suitable simply connected region, the standard cross-check is $M_y = N_x$. Then every solution satisfies $F(x,y) = C$. Exactness is a property of the chosen differential form, not merely of the slope expression after division.

Potential reconstruction without guessing the missing terms



READ THE FIGURE

The integration term after integrating M with respect to x is h(y), not simply a numerical constant.

WORKED EXAMPLE

S9 and Q17: an elliptic level curve

For $(2x+y)dx+(x+2y)dy=0$,

$$M_y = 1 = N_x.$$

Integrating M in x gives $F = x^2+xy+h(y)$. Then $F_y = x+h'(y) = x+2y$, so $h = y^2$. Hence

$$x^2+xy+y^2 = C.$$

At $(0,1)$, $C=1$. The explicit branch through that point is

$$y = \frac{-x+\sqrt{4-3x^2}}{2}, \qquad -\frac{2}{\sqrt{3}} < x < \frac{2}{\sqrt{3}}.$$

The endpoints make $x+2y=0$, so the graph cannot be continued as a finite-slope function $y(x)$ through them.

WORKED EXAMPLE

P7 and Q18: retain the x-squared-y term

For $(3x^2+2xy)dx+(x^2+2y)dy=0$,

$$M_y = 2x = N_x.$$

Integrate M:

$$F = x^3+x^2y+h(y).$$

Comparison of $F_y = x^2+h'(y)$ with $N = x^2+2y$ gives $h = y^2$. Since the initial point is $(0,1)$, the solution is

$$x^3+x^2y+y^2 = 1.$$

This implicit answer is complete and often cleaner than introducing a quartic square root.

Verification in differential form

Differentiate $F(x,y)=C$ implicitly:

$$F_x + F_y\frac{dy}{dx} = 0.$$

Where $F_y \neq 0$, this gives $y' = -M/N$. Where $F_y = 0$, use the original differential form or a local parametrization rather than dividing blindly.

COMMON TRAP

M sub-y must equal N sub-x, not M sub-x

The mixed-partial test compares the derivative of the dx coefficient with respect to y against the derivative of the dy coefficient with respect to x.

LESSON 08 / 18

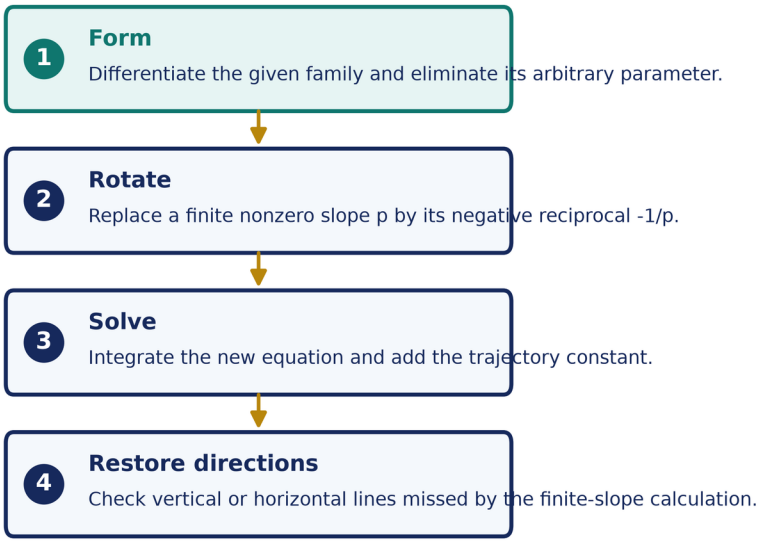
Eliminate the family parameter, then rotate the tangent

If a family satisfies $y' = p(x, y)$, an intersecting curve with finite nonzero perpendicular slope satisfies

$$y' = -\frac{1}{p(x, y)}.$$

Solve that new differential equation. Horizontal and vertical tangents need geometric handling because a finite-slope formula alone cannot express every perpendicular direction.

Orthogonality acts on the differential equation, not on the original y-formula



READ THE FIGURE
The negative reciprocal is applied after eliminating the family parameter, not to the parameter itself.

WORKED EXAMPLE

S10 and Q19: parabolas become ellipses

For $y = cx^2$ on $x \neq 0$, $c = y/x^2$ and $y' = 2cx$, so the family equation is

$$y' = \frac{2y}{x}.$$

On a branch with finite nonzero slope, an orthogonal trajectory therefore satisfies

$$\frac{dy}{dx} = -\frac{x}{2y} \quad \Longrightarrow \quad 2y\,dy + x\,dx = 0.$$

Integration gives

$$x^2 + 2y^2 = C.$$

For $C > 0$, these are ellipses centred at the origin. The calculation is local away from the origin and from points where the required slope divisions fail.

WORKED EXAMPLE

P8 and Q20: the vertical radial line must be included

For $x^2 + y^2 = c$, differentiation gives $x + yy' = 0$. Away from coordinate-axis slope degeneracies, the orthogonal equation is

$$\frac{dy}{dx} = \frac{y}{x} \quad \Longrightarrow \quad \frac{dy}{y} = \frac{dx}{x}.$$

Thus $y = Cx$ on half-plane intervals. Geometrically, every radial line is orthogonal to every concentric circle it meets. Consequently the complete nondegenerate family is

$$y = Cx, \qquad x = 0.$$

The vertical line is omitted if one reports only finite slopes; the source answer is corrected here.

A fast geometric check

Circles centred at the origin have radial normals. Their orthogonal trajectories must therefore be radial curves, namely straight lines through the same centre. For the parabolas $y = cx^2$, check an ellipse implicitly: $x + 2yy' = 0$, so its slope is $-x/(2y)$, whose product with $2y/x$ is -1 away from degenerate directions.

COMMON TRAP

An arbitrary constant is not automatically a complete geometric family

The formula $y = Cx$ excludes vertical lines. Whenever a derivation assumes a finite slope or divides by x or y, compare the result with the original geometry before calling it complete.

LESSON 09 / 18

The characteristic equation controls the solution shape

For $ay'' + by' + cy = 0$, try $y = e^{rx}$. Since each derivative multiplies by another r,

$$ar^2 + br + c = 0.$$

Two independent real solutions form the general solution. Initial values determine their constants; they do not change the characteristic roots.

Three root patterns, three independent-solution templates

Roots	Real general solution	Key point
r1, r2 real	A e^(r1 x)+B e^(r2 x)	Two rates
Repeated r	(A+Bx) e^(rx)	x gives independence
u +/- iv	e^(ux)(A cos vx+B sin vx)	Envelope + oscillation

READ THE FIGURE
For complex roots u+/-iv, v is nonzero. A repeated root without the x companion would supply only one independent solution.

WORKED EXAMPLE
P9 and Q21: pure oscillation
For $y'' + y = 0$, $r^2 + 1 = 0$, so $r = \pm i$ and

$$y = A\cos x + B\sin x.$$

The conditions $y(0) = 0$ and $y'(0) = 1$ give $A = 0$, $B = 1$. Hence $y = \sin x$ on $(-\infty, \infty)$. Here there is no exponential envelope, so this solution is genuinely periodic with period 2π.

WORKED EXAMPLE
S11 and Q22: two distinct real roots
For $y'' - 3y' + 2y = 0$,

$$r^2 - 3r + 2 = (r-1)(r-2), \qquad y = Ae^x + Be^{2x}.$$

The initial conditions give $A + B = 0$ and $A + 2B = 1$, hence $A = -1$, $B = 1$. Therefore

$$y = e^{2x} - e^x, \qquad y(1) = e^2 - e.$$

Differentiate once to verify $y'(0) = 2 - 1 = 1$.

WORKED EXAMPLE

S12 and Q23: repeated root

For $y'' + 4y' + 4y = 0$, $r = -2$ is repeated, so

$$y = (A + Bx)e^{-2x}.$$

Now $y(0) = A = 1$ and $y' = (B - 2A - 2Bx)e^{-2x}$, so $y'(0) = B - 2 = 0$ gives $B = 2$. Thus

$$y = (1 + 2x)e^{-2x}, \qquad y(1) = 3e^{-2}.$$

COMMON TRAP

Do not write two copies of the same exponential for a repeated root

The functions e^{rx} and xe^{rx} are independent; e^{rx} and another e^{rx} are not. Their Wronskian is tested in Extension 1.

LESSON 10 / 18

Variable coefficients can still produce an algebraic root equation

On $x > 0$, a second-order Cauchy–Euler equation has the form

$$ax^2y'' + bxy' + cy = 0.$$

Try $y = x^m$. Since $xy' = mx^m$ and $x^2y'' = m(m-1)x^m$,

$$am(m-1) + bm + c = 0.$$

Distinct roots m_1, m_2 give $y = Ax^{m_1} + Bx^{m_2}$. A repeated root m gives $y = x^m(A + B\ln x)$ on $x > 0$.

Constant-coefficient and Euler trials are related but not identical

Equation	Trial	Root equation or caution
Constant coefficient	exp(rx)	a r^2 + b r + c
Euler	x^m	a m(m-1) + b m + c
Repeated Euler m	x^m, x^m ln x	Log gives independence
Initial x0 > 0	Stay on x > 0	Do not cross 0

READ THE FIGURE

The Euler coefficient b multiplies m, while the coefficient a multiplies m(m-1).

WORKED EXAMPLE

S13 and Q24: roots 1 and -1

For $x^2y'' + xy' - y = 0$,

$$m(m-1) + m - 1 = m^2 - 1 = 0.$$

Thus $y = Ax + B/x$. The conditions at $x = 1$ give

$$A + B = 0, \qquad A - B = 1,$$

so $A = 1/2$, $B = -1/2$. Therefore

$$y = \tfrac{1}{2}\left(x - \tfrac{1}{x}\right), \qquad y(2) = \tfrac{3}{4},$$

on the maximal interval $(0, \infty)$.

WORKED EXAMPLE

PI2 and Q25: the initial data select a single power

For $x^2y'' - 2xy' + 2y = 0$,

$$m(m-1) - 2m + 2 = m^2 - 3m + 2 = (m-1)(m-2).$$

Hence $y = Ax + Bx^2$. At $x = 1$,

$$A + B = 1, \qquad A + 2B = 2,$$

so $B = 1$, $A = 0$. The IVP solution is $y = x^2$ and $y(3) = 9$. Extension I independently obtains the same general solution by reduction of order.

Why the logarithm appears for repeated roots

The substitution $t = \ln x$ transforms an Euler equation on $x > 0$ into a constant-coefficient equation in t. A repeated root then produces te^{mt}, and returning to x turns t into $\ln x$. This explains the template rather than memorizing it separately.

COMMON TRAP

The standard form has singular coefficients at x = 0

After dividing by x^2, the coefficients contain $1/x$ and $1/x^2$. Solutions from an initial point with $x > 0$ stay on $x > 0$ unless a separate continuation analysis is supplied.

LESSON 11 / 18

Model the rate statement before integrating

Exponential models arise when a relative rate is constant: $P'/P = k$ gives $P = P_0 e^{kx}$. Newton cooling instead says that the temperature excess above the environment decays exponentially. An autonomous second-order equation has no explicit x; the substitution $p = y'$, $y'' = p\,dp/dy$ lowers its order.

Translate each verbal or structural clue into a differential equation

Clue	Model	Consequence
Relative rate k	P' = kP	P = P0 e^(kx)
Cooling to Ta	T' = -k(T-Ta)	Excess decays
No explicit x	p = y'	y'' = p dp/dy
Divide by p or y	Nonzero branch	Test equilibria separately

READ THE FIGURE
A yearly percentage described as discrete compounding is not the same convention as a continuous relative growth rate.

WORKED EXAMPLE
S14 and Q26: cooling differences can be squared
Let T be temperature and $k > 0$. Newton's law gives

$$T - 20 = 80e^{-kx}.$$

After 10 minutes, $60 - 20 = 80e^{-10k}$, so $e^{-10k} = 1/2$. After another 10 minutes,

$$T(20) = 20 + 80(e^{-10k})^2 = 20 + 80 \cdot \tfrac{1}{4} = 40\,^\circ\mathrm{C}.$$

The temperature excess halves every 10 minutes; the absolute temperature does not.

WORKED EXAMPLE
P10 and Q27: state the continuous-growth convention
If "10% per year" means the continuous relative rate $P'/P = 0.1$, then

$$P(t) = 1000e^{0.1t}, \qquad P(10) = 1000e \approx 2718.2818.$$

This is the source's intended answer. If instead growth occurs by annual discrete compounding, the model is $P(t) = 1000(1.1)^t$ and $P(10) = 1000(1.1)^{10} \approx 2593.7425$. The modelling convention changes the answer.

WORKED EXAMPLE

S16 and Q31: reduce yy'' = (y')^2

Put $p = y'$. On a branch with $y \neq 0$ and $p \neq 0$,

$$yp\frac{dp}{dy} = p^2 \quad \Longrightarrow \quad \frac{dp}{p} = \frac{dy}{y}.$$

Thus $p = Cy$, so $y' = Cy$ and $y = Ae^{Cx}$ on a nonzero branch. The initial data $y(0) = 1$, $y'(0) = 1$ give $A = C = 1$, hence

$$y = e^x.$$

The separation also excludes constant solutions from the derivation; direct substitution shows that every constant $y = A$ solves the original equation, including $y = 0$. None changes this IVP.

COMMON TRAP

y'' is not dp/dx after choosing p(y)

When $p = y'$ is regarded as a function of y, the chain rule gives $y'' = dp/dx = (dp/dy)(dy/dx) = p\,dp/dy$.

LESSON 12 / 18

The discriminant determines the regime

For $\omega > 0$ and $\zeta \geq 0$, write

$$y'' + 2\zeta\omega y' + \omega^2 y = 0.$$

The roots are $-\zeta\omega \pm \omega\sqrt{\zeta^2 - 1}$. Negative real parts produce decay, but only complex roots produce oscillation.

Damping classification from zeta and the characteristic roots

Regime	Roots	Behaviour
Under: zeta < 1	Complex, Re < 0	Decaying oscillation
Critical: zeta = 1	Repeated negative	Repeated-root template
Over: zeta > 1	Two negative real	No sustained oscillation
Undamped: zeta = 0	Purely imaginary	Periodic, no decay

READ THE FIGURE

A damped sinusoid is not periodic: shifting by its oscillation interval also multiplies the envelope by a nontrivial exponential.

WORKED EXAMPLE

S21 and Q29: under-damped, not periodic

For $y'' + 2y' + 5y = 0$, $r = -1 \pm 2i$ and

$$y = e^{-x}(A\cos 2x + B\sin 2x).$$

The initial conditions give $A = 0$ and $2B = 2$, so

$$y = e^{-x}\sin 2x, \qquad y(1) = e^{-1}\sin 2 \approx 0.3345118292.$$

Here $\omega = \sqrt{5}$, $\zeta = 1/\sqrt{5} < 1$. Zeros occur at $x = n\pi/2$, so consecutive zeros are $\pi/2$ apart. Corresponding phases of $\sin 2x$ recur after π, the oscillation or quasi-period, but

$$y(x + \pi) = e^{-\pi}y(x) \neq y(x)$$

unless $y(x) = 0$. Thus the source's word "period" is corrected: the damped solution is not periodic.

WORKED EXAMPLE

P16 and Q30: over-damped with one initial rise

For $y'' + 3y' + 2y = 0$, the roots are -1 and -2, so the regime is over-damped. Write $y = Ae^{-x} + Be^{-2x}$. The initial data give $A + B = 0$ and $-A - 2B = 1$, hence

$$y = e^{-x} - e^{-2x} = e^{-2x}(e^{x} - 1).$$

Therefore $y > 0$ for every $x > 0$ and $y \to 0$; there is no positive zero crossing. Also $y(1) = e^{-1} - e^{-2} \approx 0.2325441579$. The maximum occurs at $x = \ln 2$, where $y = 1/4$.

Envelope versus actual graph

For Q29, the curves e^{-x} and $-e^{-x}$ bound the solution but are not the solution. The actual graph touches an envelope only where $|\sin 2x| = 1$ and crosses zero between touches. Visual Lab 4 displays all three curves with exact tick spacing.

COMMON TRAP

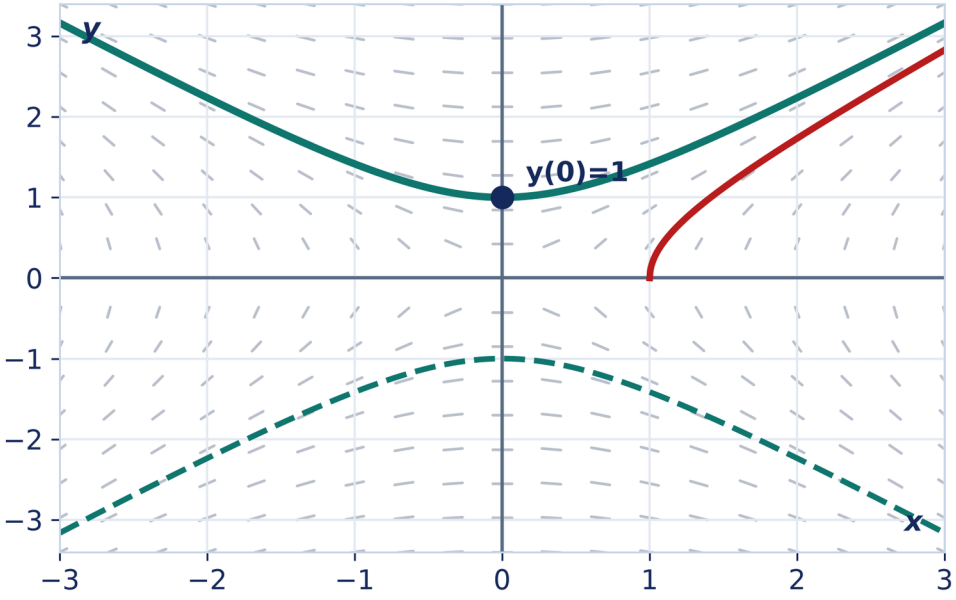
Decay alone does not imply over-damping

Under-damped and critically damped solutions can also tend to zero. Classify by the roots or by ζ, not merely by the long-term limit.

LESSON 13 / 18

Every short segment has slope x/y

The slope field for y-prime = x/y with three solution branches



READ THE FIGURE

The teal branches have y^2-x^2=1; the red branch has y^2-x^2=-1. Segments on y=0 are omitted because x/y is undefined there. The formerly cropped lower branch is now visible.

Read the geometry algebraically

Differentiate $y^2 - x^2 = C$:

$$2yy' - 2x = 0 \quad \Longrightarrow \quad y' = \frac{x}{y}.$$

Conversely, separation gives $y^2 - x^2 = C$ on any nonzero-y solution interval. The initial point $(0, 1)$ selects $C = 1$ and the positive branch $y = \sqrt{x^2 + 1}$.

WORKED EXAMPLE

Check three displayed tangent directions

At $(0, 1)$ the slope is 0, so the teal upper branch is horizontal. At $(1, \sqrt{2})$ its slope is $1/\sqrt{2}$. At $(2, \sqrt{3})$ on the red branch its slope is $2/\sqrt{3}$. Each agrees with the local line element.

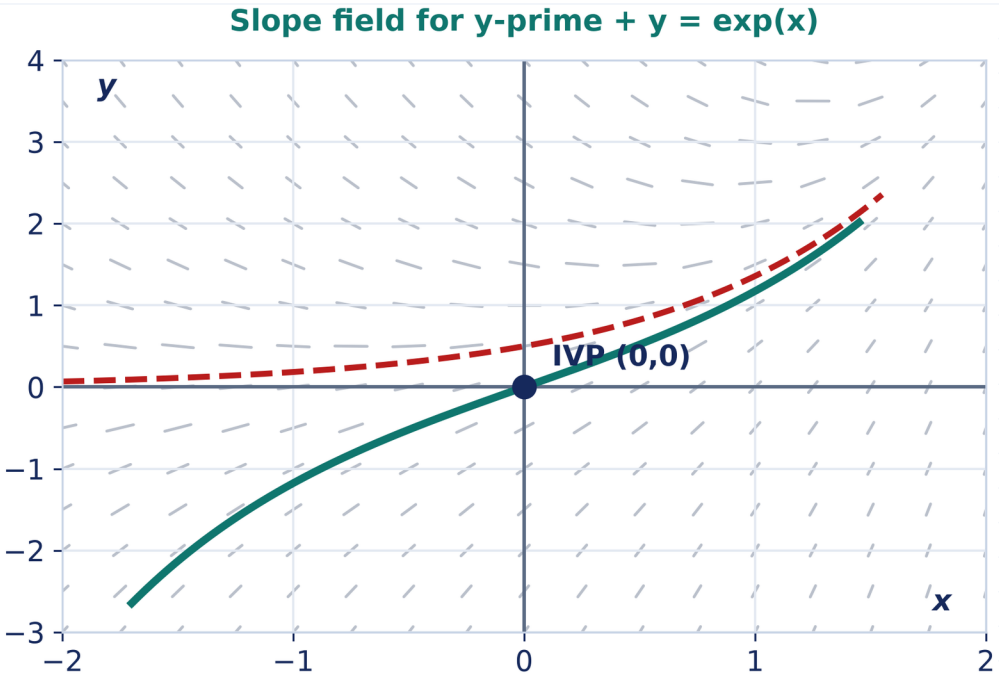
COMMON TRAP

A slope field is not a set of arrows to follow across discontinuities

The equation is undefined at $y = 0$. Do not join upper and lower branches through the x-axis merely because their formulas share the same constant C.

LESSON 14 / 18

The integrating-factor solution must follow the field



READ THE FIGURE

The solid curve is sinh x, the Q12 IVP solution. The dashed curve exp(x)/2 is the C=0 member of the same general family. Both have local slope exp(x)−y.

Read C from the initial point

The general solution is

$$y = \tfrac{1}{2}e^{x} + Ce^{-x}.$$

At $x = 0$, $y(0) = 1/2 + C$. Thus the solid IVP curve has $C = -1/2$, while the dashed curve has $C = 0$. They are distinct and cannot intersect: uniqueness for $y' = e^{x} - y$ follows because the right-hand side is smooth and has bounded y-derivative on every finite region.

WORKED EXAMPLE

Verify a tangent without the graph

For the solid curve $y = \sinh x$,

$$y' = \cosh x, \qquad e^{x} - y = e^{x} - \frac{e^{x} - e^{-x}}{2} = \frac{e^{x} + e^{-x}}{2} = \cosh x.$$

The identity holds for all real x, not only at the plotted points.

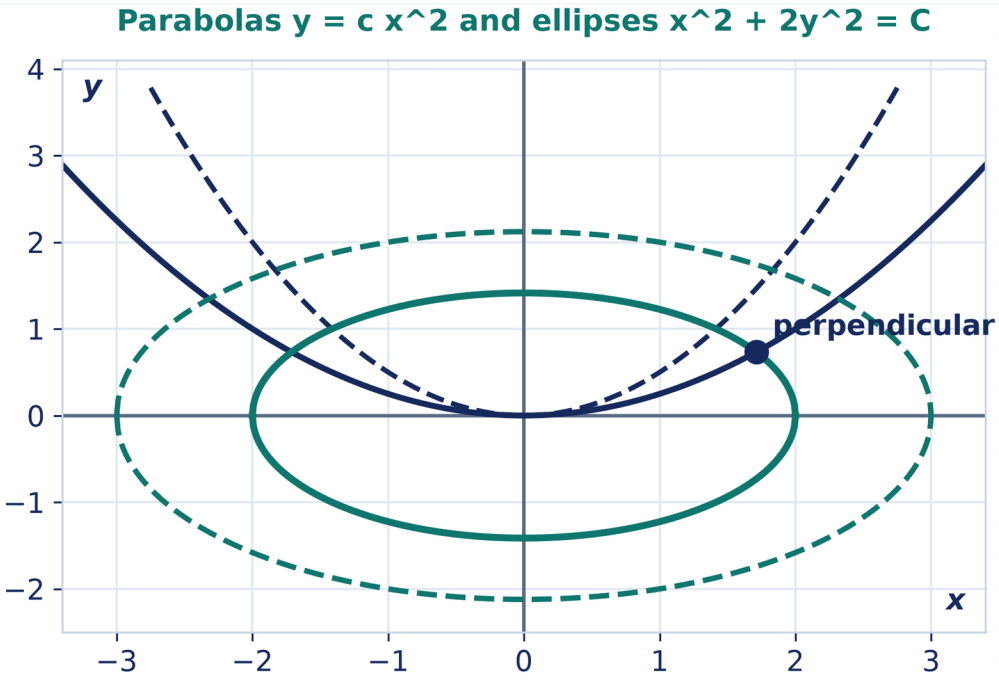
COMMON TRAP

Nearby-looking curves are not interchangeable solutions

Changing C changes the initial value. A curve can only be called the IVP solution after both the differential equation and the initial point have been checked.

LESSON 15 / 18

Two editable families recreate the orthogonal-trajectory figure



READ THE FIGURE

Navy curves are the original parabolas; teal curves are their orthogonal ellipses. Upper and lower ellipse branches are plotted separately so every equation remains editable and explicit.

Check the marked intersection exactly

The marked point lies on $y = x^2/4$ and $x^2 + 2y^2 = 4$. Put $u = x^2$. Substitution gives

$$u + \frac{u^2}{8} = 4 \quad \Longrightarrow \quad u = 4\sqrt{3} - 4.$$

At that point, the parabola slope is $x/2$, while implicit differentiation of the ellipse gives slope $-x/(2y)$. Their product is

$$\frac{x}{2} \cdot \left(-\frac{x}{2y}\right) = -\frac{x^2}{4y} = -1$$

because $y = x^2/4$ and $x \neq 0$.

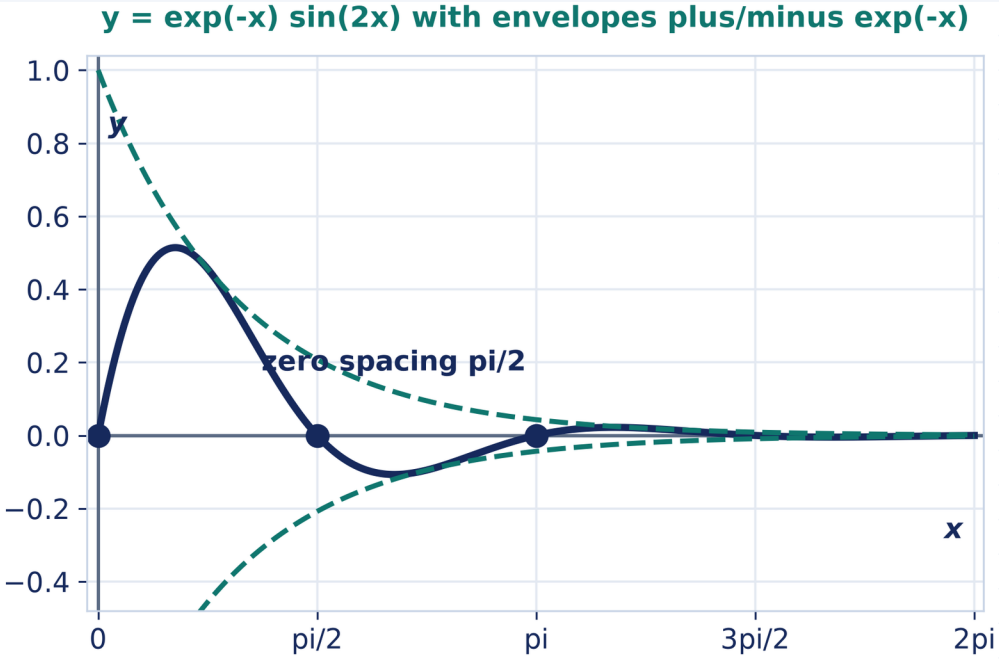
COMMON TRAP

Do not infer orthogonality from a drawing

The product-of-slopes calculation, supplemented by vertical/horizontal direction checks, is the proof. The figure only makes the two families easier to understand.

LESSON 16 / 18

Phase repeats; amplitude does not



READ THE FIGURE

The first positive peak occurs at x=(1/2) arctan(2). After pi the sine phase repeats, but every value is multiplied by exp(-pi), so the damped graph is not periodic.

Three different time intervals

- Consecutive zeros are separated by $\pi/2$.
- Corresponding sine phases are separated by π; this is the quasi-period.
- The envelope is multiplied by e^{-h} after any time shift h, and it halves when $h = \ln 2$.

WORKED EXAMPLE

Locate the first positive peak

For $y = e^{-x}\sin 2x$,

$$y' = e^{-x}(2\cos 2x - \sin 2x).$$

At the first positive maximum, $\tan 2x = 2$, so $x = \frac{1}{2}\arctan 2$, not $\pi/8$. The maximum value is obtained by substituting this exact x-value into the original formula.

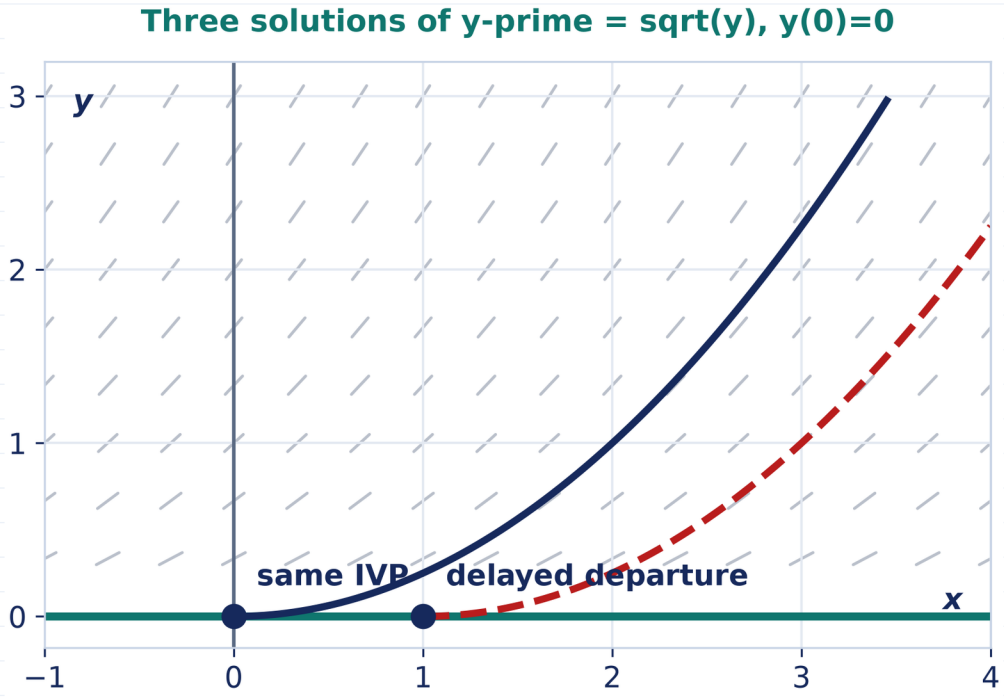
COMMON TRAP

Zeros and maxima generally have different spacing statements

Zeros of this damped sine are equally spaced, but successive maxima must be found from y-prime = 0, not guessed halfway between zeros.

LESSON 17 / 18

The zero solution can wait before peeling away



READ THE FIGURE
Teal is the zero solution; navy departs at 0; red remains zero until x=1 and then follows (x-1)^2/4. Replacing 1 by any nonnegative delay gives another solution.

Differentiability includes the joining point

For delay $a \geq 0$, the departing formula is $y = (x-a)^2/4$ for $x > a$ and $y = 0$ for $x \leq a$. On the right,

$$y' = \frac{x-a}{2} = \sqrt{y}.$$

At $x = a$, both one-sided derivatives are 0, and the right-hand side is also 0. Hence the joined function solves the IVP on the whole real line.

WORKED EXAMPLE

Why the uniqueness theorem does not object

For $f(y) = \sqrt{y}$ and $y > 0$,

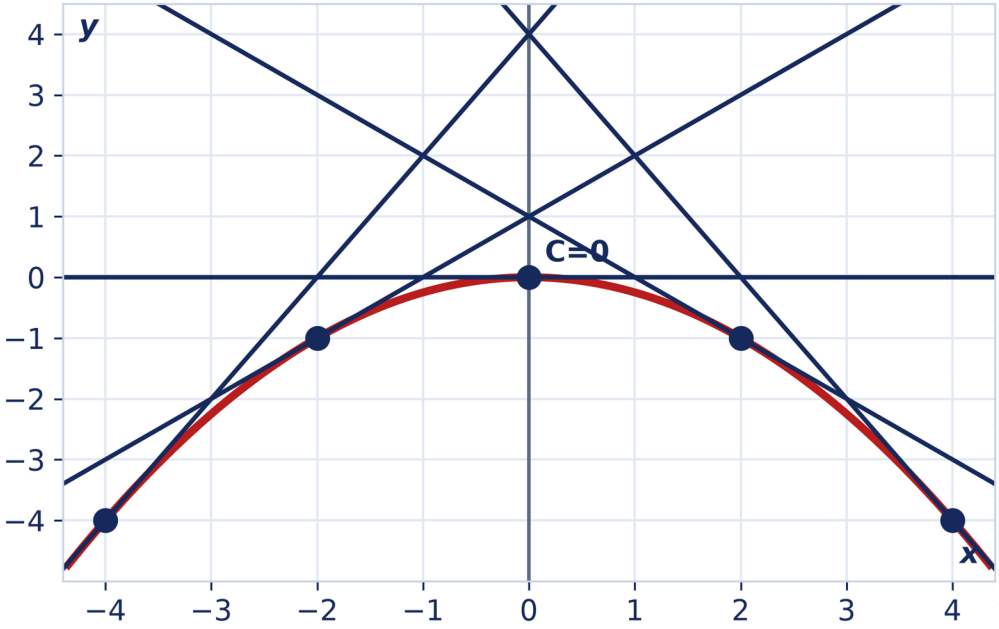
$$f_y = \frac{1}{2\sqrt{y}},$$

which is unbounded as $y \to 0^+$. Therefore f is not Lipschitz on a neighbourhood of $y = 0$. Continuity still permits existence; the missing Lipschitz hypothesis removes the uniqueness guarantee. The negative plotting margin is only visual space: real $\sqrt{y}$ requires $y \geq 0$.

LESSON 18 / 18

The singular solution is touched by every straight-line member

Lines y = Cx + C^2 and the envelope y = -x^2/4



READ THE FIGURE

For each C, the line touches the red parabola at x=-2C and y=-C^2. The line and envelope have the same value and derivative there.

Eliminate C by envelope conditions

A line satisfies $F(x, y, C) = y - Cx - C^2 = 0$. Its envelope also satisfies

$$F_C = -x - 2C = 0.$$

Thus $C = -x/2$ and

$$y = Cx + C^2 = -\frac{x^2}{2} + \frac{x^2}{4} = -\frac{x^2}{4}.$$

Direct differentiation gives $y' = -x/2 = C$ at the contact point, confirming both value and slope agreement.

WORKED EXAMPLE

The envelope also solves the differential equation

If $y = -x^2/4$, then $p = y' = -x/2$ and

$$xp + p^2 = -\frac{x^2}{2} + \frac{x^2}{4} = -\frac{x^2}{4} = y.$$

This is why the envelope is a genuine singular solution, not merely a visual boundary. No finite C produces the parabola; it is an additional solution and must be listed separately.